V_{ud} at next-to-next-to-leading order in the electroweak interactions

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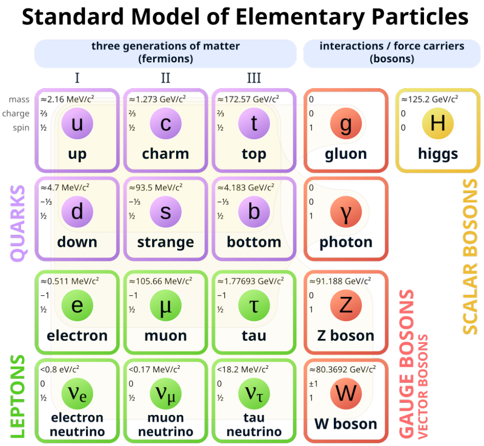
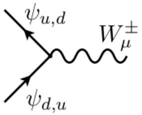


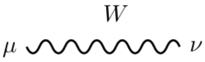
Figure 1: Standard Model

Weak decays

We are interested in charged interactions - mediated by the W bosons.



$$i\frac{g}{\sqrt{2}}\gamma_\mu\frac{1-\gamma_5}{2}$$



$$\frac{-ig_{\mu\nu}}{k^2-M_W^2+i\epsilon}$$

Figure 2: Feynman rules for the Weak Interaction

- Leptonic decays are decays that involve only leptons and neutrinos;
- Semi-leptonic decays are decays that involve quarks and leptons.

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

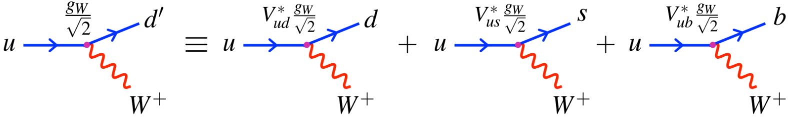


Figure 3: CKM matrix

- CKM matrix elements are expected complex constants
- CKM matrix is expected to be unitary: $V^{\dagger}_{CKM}V_{CKM} = 1$

The current experimental values for the CKM, from [?]

$$\begin{pmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{pmatrix} =$$

$$\begin{pmatrix} 0.97373 \pm 0.00031 & 0.2243 \pm 0.0005 & 0.00394 \pm 0.00036 \\ 0.221 \pm 0.004 & 0.987 \pm 0.011 & 0.0422 \pm 0.0008 \\ 0.0081 \pm 0.0005 & 0.0394 \pm 0.0023 & 0.9991 \pm 0.0001 \end{pmatrix}$$

The CKM matrix is expected to be unitary in the first row,

$$\Delta_{CKM} = 1 - |V_{ud}|^2 - |V_{us}|^2 - O|V_{ub}|^2 = 0 \tag{1}$$

where $|V_{ub}|^2 \approx 1.5 \times 10^{-5}$

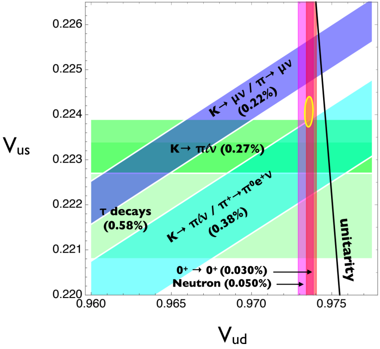


Figure 4: Summary of constraints on V_{ud} and V_{us} (assuming the Standard Model hypothesis) from nuclear, nucleon, meson, and τ lepton decays., [?].

Motivation

- Compute 2 loop EW corrections relevant for a pion decay;
- Match the amplitudes in the full theory to the EFT, to extract the Wilson coefficients (short distance contribution);
- WC contribute to later extract the CKM element V_{ud};
- Compute the ratio of the pion decay normalized to the muon decay (therefore we compute as well 2 loop EW corrections for the muon decay)
- Check that if the pion decay normalized to the muon decay, the only contributions to the WC come from box-type Feynman diagrams.

Tree-level Feynman diagrams in the SM and EFT

Figure 5: SM tree-level.

Figure 6: EFT Tree-level.

Evaluating the amplitude A_{tree}:

$$A_{tree} = \frac{-G_F}{2} V_{ud} \frac{g^{\mu\nu}}{k^2 - m_W^2} [\bar{d}\gamma_\mu P_L u][\overline{\nu}\gamma_\nu P_L l] \tag{2}$$

but if we consider small internal momenta $k << m_W \approx 80 GeV$ we can expand the amplitude. The leading term is given by

$$A = \frac{-G_F}{2} V_{ud} \frac{-1}{m_W^2} [\bar{d}\gamma_\mu P_L u][\overline{\nu}\gamma_\mu P_L l] = \frac{-G_F}{2} V_{ud} \frac{-1}{m_W^2} [\bar{d}u]_{V-A}[\overline{\nu}l]_{V-A} \tag{3}$$

Basic idea of OPE: the product of two charged current operators is expanded into a series of local operators, whose contributions are weighted by effective coupling constants, the Wilson coefficients. The **effective weak Hamiltonian** has the following structure

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM}^i C_i(\mu) Q_i(x) \tag{4}$$

where G_F is the **Fermi constant**
V_{CKM}^i are the **CKM matrix elements**
C_i are the **Wilson coefficients** $\longrightarrow$ short distance behavior
Q_i are the **local operators** $\longrightarrow$ long distance behavior

Physical and Evanescent operators

$$Q_1 = (\bar{d}\gamma^\mu P_L u)(\bar{\nu}\gamma^\mu P_L e)$$

$$Q_1^\mu = (\bar{\nu_\mu}\gamma^\mu P_L \mu)(\bar{\nu_e}\gamma^\mu P_L e)$$

$$E_1 = (\overline{d}\gamma^\mu\gamma^\nu\gamma^\beta P_L u)(\overline{\nu}\gamma_\mu\gamma_\nu\gamma_\beta P_L e) - (16 - a_q^1\epsilon)Q_1$$

$$E_1^\mu = (\overline{\nu_\mu}\gamma^\mu\gamma^\nu\gamma^\beta P_L \mu)(\overline{e}\gamma_\mu\gamma_\nu\gamma_\beta P_L e) - (16 - a_\mu^1\epsilon)Q_1^\mu$$

$$E_2 = (\overline{d}\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\gamma^\sigma P_L u)(\overline{\nu}\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta\gamma_\sigma P_L e) - (256 - a_q^2\epsilon)Q_1 + 20E_1$$

$$E_2^\mu = (\overline{\nu_\mu}\gamma^\mu\gamma^\nu\gamma^\alpha\gamma^\beta\gamma^\sigma P_L \mu)(\overline{\nu_e}\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta\gamma_\sigma P_L e) - (256 - a_\mu^2\epsilon)Q_1 + 20E_1$$

1-loop EFT

The 1-loop EFT diagrams relevant for the pion decay

Figure 7: 1-loop EW correction to the EFT theory with radiative corrections. The curved lines represent photons.

Choice of kinematics: external momenta set to zero and all masses bellow the W mass are set to zero. Only masses present in our calculation will be m_W, m_Z, m_H, m_t.

Exact decomposition of a propagator

$$\frac{1}{(k+p)^2-m^2} = \frac{1}{k^2-M^2} + \frac{-p^2+2k\cdot p-m^2+M^2}{k^2-M^2}\frac{1}{(k+p)^2-m^2} \tag{5}$$

k and p stand for a linear combination of the internal moment and external momenta. The propagator mass is m and M is an introduced fictitious regulator.

The sum of the amplitudes of the 1-loop diagrams of the EFT

$$M_{EFT} = \int \frac{d^d k}{(4\pi)^d} \frac{1}{(k^2 - M^2)^2} (a_1 Q_1 + a_2 E_1) \tag{6}$$

with a_1 and a_2 being functions dependent on G_F, the charges of the particles, the dimension d and the photon gauge ξ.

Feynman integral $\longrightarrow$ regularise the divergences $\longrightarrow$ introduce parameter that isolates the divergences

DimReg $\longrightarrow$ space-time dimension is extended to $4 \longrightarrow d = 4 - 2\epsilon$. The singularities are the poles when $\epsilon \longrightarrow 0$

$$\textit{2-loop:}\frac{a_1}{\epsilon^2} + \frac{b_1}{\epsilon} + c_1 \tag{7}$$

with a_1, b_1 and c_1 being constants.

The bare EFT amplitude at 1 loop is:

$$M_{EFT} = \frac{\alpha}{4\pi} G_F \frac{1}{\epsilon}(-2Q_1 - \frac{1}{12}E_1) \tag{8}$$

1-loop SM amplitudes

Generate the diagrams with the SM

$\downarrow$

Compute the amplitudes $\longrightarrow$ tensor integrals $\longrightarrow$ tensor decomposition $\longrightarrow$ reduction to masters

$\downarrow$

analytical amplitudes

$\downarrow$

$\overline{MS}$ renormalization $\longrightarrow$ renormalized amplitude

1 PI amplitudes in the full SM

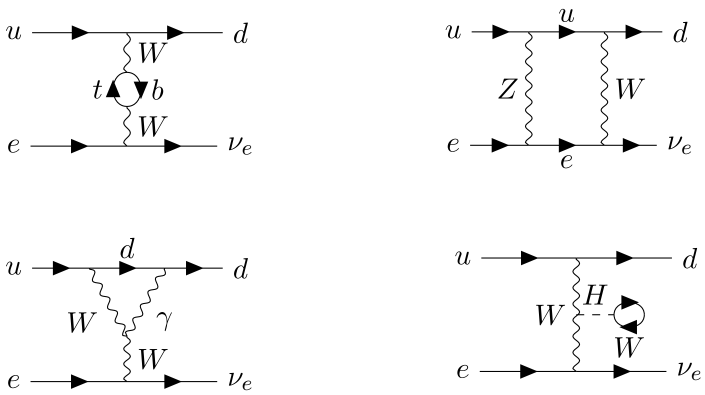


Figure 8: Some 1 loop diagrams in the full theory for the semi-leptonic decay.

The 1 loop counterterms needed are

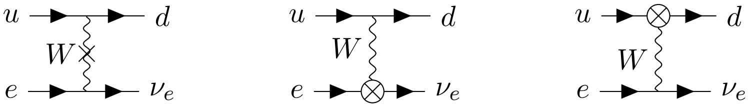


Figure 9: 1 loop counterterms in the full theory for the semi-leptonic decay.

In the $\overline{MS}$ scheme we need following renormalization constants

$$Z_{M_W}, Z_{M_Z}, Z_{s_W}, Z_e, Z_W, Z_u, Z_d, Z_{ele}, Z_\mu, Z_\nu$$

Joining all the amplitudes for all type diagrams we obtain the following for the part proportional to the physical operator Q_1,

$$M_{SM} = G_F(\frac{\alpha}{4\pi}(a_1 + a_2 \ln\left(\frac{m_t^2}{m_W^2}\right) + a_3 \ln\left(\frac{M_H^2}{m_W^2}\right) + a_4 \ln\left(\frac{M_Z^2}{m_W^2}\right) + a_5 \ln\left(\frac{\mu^2}{m_W^2}\right)) + \frac{\alpha}{4\pi}\frac{(-2)}{\epsilon}) \tag{9}$$

Notice: we performed renormalization of all physical quantities, but we still have a divergence in the amplitude in the SM.

We have the amplitudes now renormalized, so we can do the matching. Let us compute C_1 proportional to the physical operator Q_1.

The Wilson coefficients can be extracting by imposing that the full amplitude (proportional to the physical operator Q_1) should match the EFT amplitude (proportional to Q_1).

$$M_{full} = M_{EFT} \tag{10}$$

$$C_1 = \frac{\alpha}{4\pi}(a_1 + a_2 \ln\left(\frac{m_t^2}{m_W^2}\right) + a_3 \ln\left(\frac{M_H^2}{m_W^2}\right) + a_4 \ln\left(\frac{M_Z^2}{m_W^2}\right) + a_5 \ln\left(\frac{\mu^2}{m_W^2}\right)) \tag{11}$$

Wilson Coefficients in the ratio of the decays

We now do the ratio of 2 effective theories in perturbation theory. The Lagrangian is

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q \tag{12}$$

And the Wilson Coefficients are

$$C_r^{LO} = 1 \tag{13}$$

$$C_r^{NLO} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2\log\left(\frac{\mu^2}{M_Z^2}\right). \tag{14}$$

- By analyzing this result we can verify that it comes solely from the box diagrams - all the other topologies cancel.

2 loop

We have generated all the 2 loop order all the box-type diagrams for both the pion and muon decay. The unrenormalised amplitudes were simplified to a final analytical form.

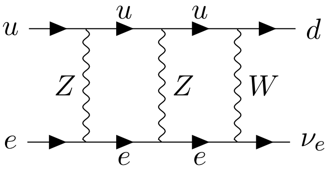


Figure 10: Example of a 2 loop diagram for the pion decay.

The counterterms at 2 loop were generated to a final analytical form.

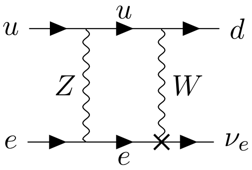


Figure 11: Example of counterterms for the 2 loop renormalization

2 loop calculation

- We are currently studying the cancellations between diagram types that occurs at 2 loop and generating the 2 loop renormalization constants for external particles.
- As our final goal, we aim to compute the 2 loop Wilson coefficients in the pion decay normalized to the muon decay and understand which diagrams contribute to this.

Conclusions

1 loop

- We were able to study how to renormalize the SM and EFT amplitudes to be able to match them
- We computed the Wilson coefficients of each decay and checked they were finite
- We computed the pion decay normalized to the muon decay, showing that the only contributions that survive com from box-type diagrams

2 loop

- We generated the analytical result of the bare amplitudes for the pion and muon decay.
- We are generating the 2 loop counterterms and respective renormalization constants
- We are working on checking which cancellations occur between diagrams when computed the 2 loop pion decay normalized to the muon decay.